Where You Actually Feel It

Structural Foundations of Health — Part 2 of 2

Last class, in one slide

Without looking at your notes: what does it mean that health is a *staircase* and not a *cliff*?

Turn to your neighbour. Thirty seconds.

Retrieval practice, not review. Take two answers, correct gently, move on. Target answer: health improves at every step up, so it is not just a poverty problem.

This lecture

Advantage is distributed unevenly, and the distribution has a history.

Home

Work

Neighbourhood

Policy

H**ow you would study a real community**?

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1. Home

Housing does three separate things

- **What it is made of** — lead, mould, pests, cold
- **What it costs you** — rent takes the money that would have been food or a co-pay
- **Whether you get to stay** — three moves in a year: new school, lost doctor, no sleep

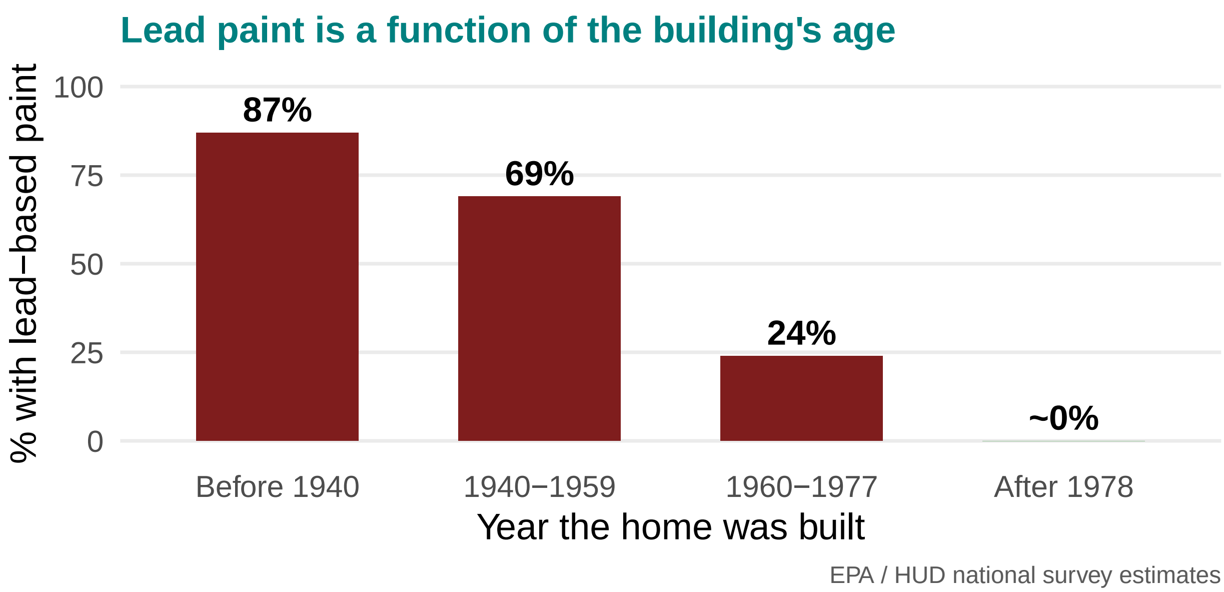
Only the first is about the building. The other two are **money** and **stability**.

One concrete case: lead

Lead paint was legal in US homes until **1978** — and is still in those walls.

Old housing is concentrated in poor neighbourhoods. Lead damages the developing brain, and **the damage does not reverse**.

Where the lead actually is

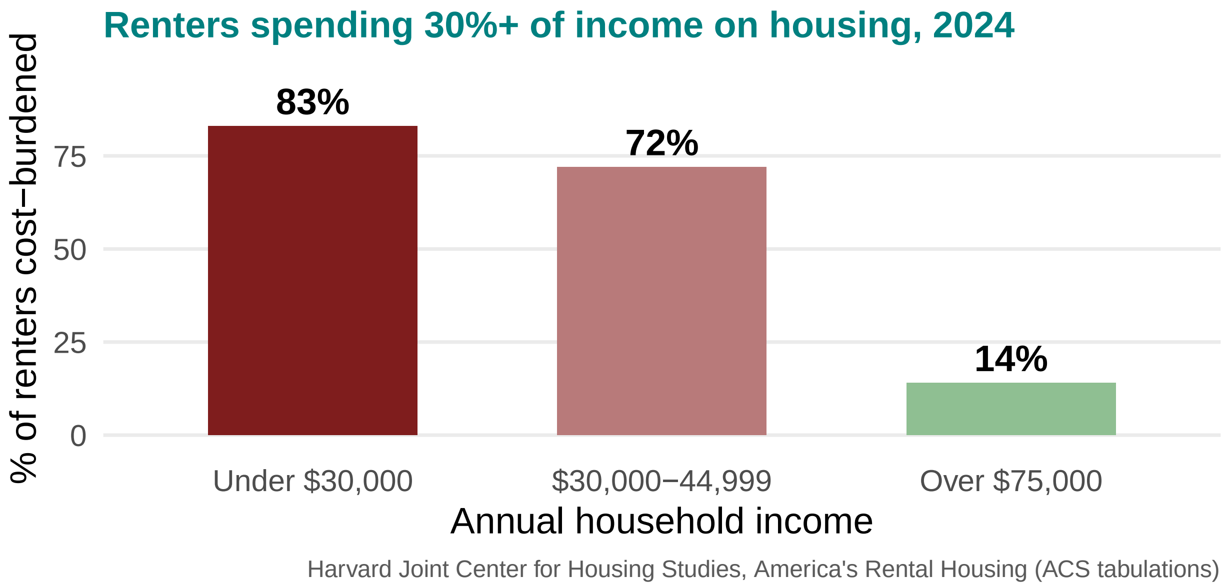


Old housing is not scattered at random. **It is where the D grades were.**

And the cost side

Paying more than **30%** of income on rent = **cost-burdened**. Above 50% = **severely** cost-burdened.

Who pays too much



The staircase again — this time in rent. Not a cliff at the poverty line.

So what gives?

Quick one: rent goes up $200 a month, income doesn't. What comes out of the budget first?

Name three things. Which of them are health?

Expected: groceries, prescriptions, the dentist, the car repair that gets you to work, heating. All of them are health. This makes housing cost a health variable, not a finance variable.

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2. Work

The wrong question and the right one

Wrong question: **do you have a job?**

Right question: **what does the job do to you?**

Two people both “employed” can have completely different health prospects.

Three ways a job reaches your body

The body takes it

Lifting, repetition, chemicals, heat

The mind takes it

High demand, low control, unpredictable hours

It follows you home

Pesticides, solvents, dust on your clothes

The middle one is the surprise: **high demand + low control** predicts heart disease. Being busy is not the problem. Being busy with no say is.

The middle one, drawn

← low control   high control →

Passive

Low demand, low control

Night watch, some clerical work. Skills erode; disengagement.

Low strain

Low demand, high control

Self-paced, autonomous. The healthiest quadrant.

High strain

High demand, low control

Line work, call centre, care aide, warehouse pick rates. Highest cardiovascular risk.

Active

High demand, high control

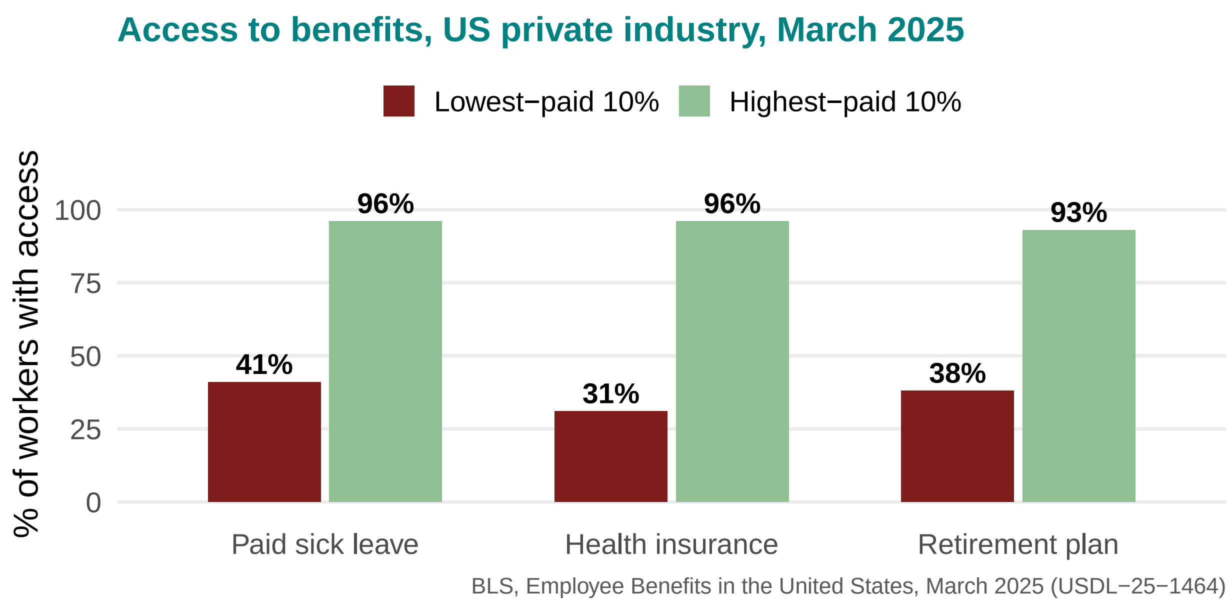
Surgeon, senior manager. Demanding — but not, on its own, damaging.

← low demand   high demand →

Two people can work equally hard. Only one of them has a say in how.

Karasek's demand-control model (1979). The quadrant students always want to argue about is Active: yes, the surgeon's job is brutal, and no, it does not carry the same risk, because the surgeon decides the order of the day. Ask them which quadrant the jobs they have actually held fall into — it is usually the dark one, and that lands better than any statistic.

Who actually gets benefits



"Access" means the employer offers it, not that the worker takes it up — so these bars are the optimistic version.

Why paid sick leave is a health policy

41% of the lowest-paid workers can take a paid sick day. **96%** of the highest-paid can.

So the person handling your food, minding your child, or turning your grandmother in bed is the one who **cannot afford to stay home sick**.

Their working conditions are **your** exposure. This is why we study populations, not individuals.

This is the slide that makes the population-level argument land for students who came in thinking public health is about personal responsibility.

And a growing group with none of it

Independent contractors: **7.4%** of US employment — 11.9 million people.

Health insurance from *any* source:

- Traditional employees — **85%**

- Independent contractors — **74%**
- Temp agency workers — **61%**

No employer means no employer-sponsored anything: no sick leave, no workers' comp, no unemployment insurance. **Insecure work, insecure health.**

Source: BLS Contingent and Alternative Employment Arrangements, July 2023 (released Nov 2024). Note honestly that BLS does not publish an employer-coverage figure for contractors, because by definition there is no employer.

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3. Neighbourhood

What is built around you

Ask what they can see: sidewalks, crossings, lighting, shade, benches, whether the shop is walkable. The point is that "exercise" is a design outcome, not only a willpower outcome.

And what is put near you

Environmental justice: pollution, heat and industrial hazards are not spread evenly — they concentrate in low-income neighbourhoods and neighbourhoods of colour.

Remember last class: **cheap land is where the highway goes.**

Which came first — the polluting plant, or the poor neighbourhood around it?

There is evidence for both. Why does the answer matter for policy?

Genuinely contested in the literature (siting vs. subsequent demographic change). Use it to show that "which came first" changes whether the remedy is siting rules or cleanup and relocation.

What it does

- **Lungs** — asthma, COPD; children worst affected
- **Heart** — fine particulate exposure raises cardiovascular risk
- **Heat** — less shade, more asphalt; heatwave deaths cluster in the same blocks
- **Mind** — noise, no green space, and knowing this was allowed to happen here

This is the redlining map again, wearing a different coat.

And what happens on the street

Photo: Associated Press

SAY SOMETHING BEFORE THIS SLIDE. One sentence is enough: "The next few slides are about firearm injury. Some of you will have lost someone. Step out if you need to." Then move at a normal pace – do not linger on the image, and do not editorialise over it. The photo is here to be counted, not felt.

Every yellow marker is a piece of evidence. Do not name the city; the point is not this street.

The wrong question, one more time

Wrong question: **what is wrong with the people on that street?**

Right question: **why that street?**

Firearm injury is a leading cause of death for US children and adolescents. It is also, like everything else in this lecture, **unevenly distributed** — and the unevenness is very fine-grained.

Deliberate echo of the Work section's "wrong question / right question" slide. Students should recognise the move.

How fine-grained

under 5%

Of street segments and intersections in one large US city accounted for the **majority of its gun violence** — across 29 years.

A problem that concentrates on 5% of streets, and *stays* on those streets for three decades, is a problem about **streets**.

Braga, Papachristos & Hureau (2010), Boston, 1980-2008. The stability is the surprising part, not the concentration – the same corners, decade after decade, through a citywide crime drop. That persistence is what makes it look structural rather than incidental. Do not let this slide become a policing conversation; the next slide is the reason.

Somebody tested that

A US city had thousands of blighted vacant lots. Researchers took **541** of them, grouped them into 110 clusters, and **randomly assigned** each cluster to:

Greened

Trash cleared, graded, grass and trees planted, low fence

Cleaned only

Mowed, trash removed, nothing planted

Nothing

Left as they were

Then they counted police-recorded shootings around each lot for two years.

Flag the design out loud: this is a randomised controlled trial of a *place*, not of a drug and not of a person. Students have never seen one. It matters for the next section – this is what a population-scale answer looks like when somebody actually ran it.

What grass did

−29%

Gun violence around the treated lots, in neighbourhoods **below the poverty line**.

Nearby residents also reported feeling less depressed — by more than **68%** in those same neighbourhoods.

The intervention was grass, a fence, and a mower. **No policing. No programme. Nobody's behaviour was corrected.**

Branas et al. (2018), PNAS; depression finding from South et al. (2018), JAMA Network Open. Be straight about the limits, because they will ask: the citywide effect was much smaller (about -5.8%), and -29.1% is the below-poverty-line subgroup. Say that out loud. The honest framing is that the effect is concentrated exactly where the blight is, which is the argument, not a caveat.

If you want one line to close the section: the vacant lot was a decision too. Somebody stopped maintaining it, and somebody could start.

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4. Policy

The lever

Home, work and neighbourhood are all **downstream of decisions somebody made**.

Pushes health up

- Income support (EITC)
- Medicaid expansion
- Paid leave laws
- Lead abatement funding

Pushes health down

- Zoning that blocks affordable housing
- Mass incarceration
- Weak housing-code enforcement

None of these is called a health policy. All of them are.

A natural experiment: Medicaid expansion

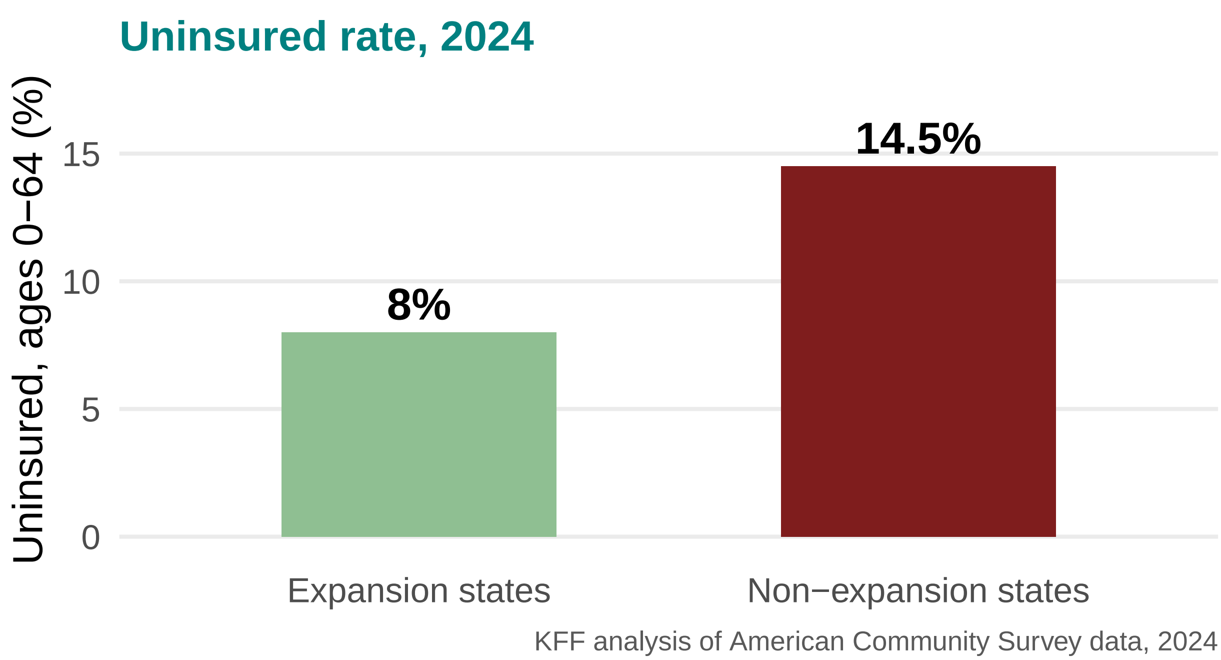
In 2014, some states expanded Medicaid and some did not. Same country, same decade.

9.4%

Reduction in annual mortality among low-income adults aged 55–64 in expansion states.

Miller, Johnson & Wherry (2021), Quarterly Journal of Economics. Be precise about the population: low-income adults aged 55–64, not everybody. Rising to 11.9% by the third year. Roughly 15,600 deaths would have been averted had all states expanded.

And the rest of the picture



Expansion also cut medical debt sent to collections by about **42%** (Hu et al. 2018).

Flag the honest caveat: the uninsured bars are a cross-section, not a causal estimate — expansion and non-expansion states differ in many ways. The mortality and debt figures are the causal ones.

How would you study a community?

Two different jobs

Community assessment

One place, in depth.

- You go there; you talk to people
- Assets as well as problems
- Ends with: a plan somebody can act on

Population assessment

Many places, from above.

- Registries, surveys, vital statistics
- Compare counties, states, countries
- Ends with: a pattern, and where to look next

Neither is better. Each is bad at the other's question.

Same city, two questions

- **Population:** *Which PA counties have the highest diabetes rates?* → surveillance data, a map, a ranking
- **Community:** *Why this neighbourhood, and what already exists here that could help?* → walk the streets, count the shops, ask the clinic

Your individual paper is a community assessment — of your own hometown. That is why it cannot be written from data alone.

Deliberate hook to the individual paper due 24 November. Point out that the observation part is what earns the marks.